

Skills Summary

Identities, Multiple Angles and Exact Solution Sets

Use this reference while completing your worksheet or when studying.

Skill 1 — Pythagorean identities

$$\sin^2 \theta + \cos^2 \theta = 1, \quad \text{so} \quad \sin^2 \theta = 1 - \cos^2 \theta \quad \text{and} \quad \cos^2 \theta = 1 - \sin^2 \theta.$$

$$\text{Also } 1 + \tan^2 \theta = \sec^2 \theta \quad \text{and} \quad \tan \theta = \frac{\sin \theta}{\cos \theta}.$$

Use these to turn a mixed expression into one written in a single ratio.

Example: Simplify $\frac{1 - \sin^2 \theta}{\cos \theta}$.

Step 1. The numerator is one of the Pythagorean rearrangements, so replacing it first turns the fraction into a cancellation.

Step 2. $1 - \sin^2 \theta = \cos^2 \theta$, so the fraction is $\frac{\cos^2 \theta}{\cos \theta} \Rightarrow \cos \theta$

Try it: Simplify $\tan \theta \cos \theta$.

check: $\sin \theta$

Skill 2 — Sum and difference formulas

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B \quad \cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

Watch the sign on cosine: it *flips*. Build an unfamiliar angle out of 30° , 45° , 60° and 90° .

Example: Find the exact value of $\sin 75^\circ$.

Step 1. Seventy-five is not a special angle, but it is a sum of two that are, so a sum formula turns it into known values.

$$\text{Step 2. } \sin(45^\circ + 30^\circ) = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} \Rightarrow \sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$$

Try it: Find the exact value of $\cos 75^\circ$.

check: $\frac{\sqrt{6} - \sqrt{2}}{4}$

Skill 3 — Double-angle formulas

$$\sin 2\theta = 2 \sin \theta \cos \theta \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta = 1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1$$

Choose the version of $\cos 2\theta$ written in the ratio the rest of the problem already uses — that is what leaves a quadratic in one function.

Example: If $\sin \theta = \frac{1}{3}$, find $\cos 2\theta$.

Step 1. Only the sine is known, so pick the form of the double angle written in sine alone and no cosine is needed at all.

Step 2. $\cos 2\theta = 1 - 2 \sin^2 \theta = 1 - 2 \left(\frac{1}{9} \right) \Rightarrow \frac{7}{9}$

Try it: If $\sin \theta = \frac{2}{5}$, find $\cos 2\theta$.

check: $\frac{25}{17}$

Skill 4 — Exact solution sets

Method: reduce to one ratio, factor, set each factor to zero, and place each value in every quadrant its sign allows.

Count check: a value strictly between -1 and 1 gives *two* angles per turn; the extremes 1 and -1 give exactly *one*.

Example: Solve $(2 \sin \theta - 1)(\sin \theta + 1) = 0$ for $0^\circ \leq \theta < 360^\circ$.

Step 1. A product is zero when a factor is zero, so this is two small equations, and the count check tells me what to expect from each.

Step 2. $\sin \theta = \frac{1}{2}$ gives two angles; $\sin \theta = -1$ is the minimum and gives one. $\Rightarrow \theta = 30^\circ, 150^\circ, 270^\circ$.

Try it: Solve $(2 \cos \theta + 1)(\cos \theta - 1) = 0$ for $0^\circ \leq \theta < 360^\circ$.

check: $\theta = 0^\circ, 120^\circ, 240^\circ$